WPR 1 Practice — Answer Key

Solutions, scoring rubrics, and session run plan

Instructor copy. Do not distribute before the timed run. 170 points across five parts, 105 minutes.

Aligned to HW1 and HW2. Conventions follow the course homeworks exactly: the outlier screen is $|z| > 2.5$ (HW1 Q16), residual plots are drawn against **fitted values** (HW2 Q11), KNN is a **classification** method with a majority vote (HW2 §5), and the starting k is $\sqrt{n_{train}}$ rounded to the nearest odd integer (HW2 Q22).

Answer key at a glance

Item	Answer	Item	Answer
6. Mean / median	185 / 177.5 min	17b. Training MAE	4.00 min
7. s	21.75 min	18. Residual pattern	U-shaped → linearity fails
8. Q1 / Q3 / IQR / range	173.5 / 187 / 13.5 / 68	19. Residual mean / sd	0.00 / 4.65 min
9a. z(236) / z(168)	2.34 / −0.78	20. Test MAE	7.20 min
9b. |z| > 2.5 screen	**Neither is flagged**	22. mileage mean / s	20 / 8 mi/wk
10. Measure of center	Median (right-skewed)	22b. Q standardized	(1.25, −1.00)
14. Line of best fit	$\hat{y} = 210 - 1.2x$	23. Neighbor order	E, F, D, B, C, A
15a. Slope	−1.2 min per mi/wk	24. k = 1 / 3 / 5	GO / GO / NO-GO
16. R^2	85.4% of variation	25. Accuracy 1 / 3 / 5	75% / 100% / 50%

The seven things that actually cost points

1 — Dividing by n. Sample standard deviation divides by $n - 1$. On Q7 that is 21.75 versus 20.34, and every z-score downstream is then wrong.

2 — Flagging 236 as an outlier. Q9 is a trap. z = 2.34 does **not** clear the course's |z| > 2.5 screen. Cadets who pattern-match “biggest number = outlier” lose it.

3 — Interpretation with no context or units. “The slope is −1.2” is not an interpretation. Every interpretation item requires the number, both units, the direction, and association rather than causation language.

4 — Reading residual size instead of residual pattern. Small residuals do not make a model correct. The U-shape in Q18 is the finding. Note the plot is against **fitted values**, per HW2.

5 — Treating training error as the model's accuracy. Q20 and Q26 both turn on this. A training MAE of 0.00 is a warning sign, not a result.

6 — Averaging in KNN. KNN here is **classification**: the prediction is the majority class of the k neighbors, not an average of their values. Cadets who averaged the ruck times are answering a question

the course does not ask.

7 — Standardizing the test set with test statistics. Q21(b). The training mean and standard deviation define the scale; recomputing on the test set moves the yardstick and leaks test information into the model.

Part I — Modeling Framework and Data

24 POINTS

1. Descriptive or predictive? (4 pts)

(a) **Descriptive** (b) **Predictive** (c) **Descriptive**

(b) is predictive because it asks for an estimate of an *unobserved* outcome for a cadet who is not in the dataset. Descriptive questions summarize data already in hand; predictive questions produce a value for a new case.

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1 pt per label, 1 pt justification. No credit for "it predicts something" without reference to a new or unobserved case.

2. Population, sample, and data dictionary (7 pts)

(a) The **population** is all Company B cadets who will run *this term's* 12-mile march (the group the commander must decide about). The 64 records are a **sample** — and not even a random one: they are last term's cadets, a different group in a different term. That matters because every inference carries the assumption that last term's cadets resemble this term's.

(b) **Platoon** — categorical (nominal), no units. **Ruck time** — numerical, **continuous**, minutes. **Prior ruck marches** — numerical, **discrete** (a count), marches. **Met the standard** — categorical (binary), no units.

(c) A data dictionary fixes the meaning, units, and allowed values of every column, so two analysts working the same file compute the same thing and nobody silently treats minutes as hours or averages a platoon number.

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2 pts (a) — must identify the data as a sample AND say why that matters. 4 pts (b), 1 per row. 1 pt (c). Deduct if "prior marches" is called continuous; counts are discrete.

3. Training and test sets (5 pts)

The reported error is **training error**: it measures how well the model reproduces the data used to build it, not how well it predicts a cadet it has never seen. A flexible enough model drives training error to zero and can still be useless out of sample. The cadet should have **partitioned into a training set and a test set** (roughly 80/20), fit on the training set only, and reported performance on the held-out test set — the only honest estimate of performance on this term's cadets.

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2 pts identifies in-sample / training error. 2 pts prescribes partition, fit on training, evaluate on held-out test. 1 pt connects test error to the real decision.

4. Unusual or erroneous? (4 pts)

Record 41 (34 min) is a data error. 34 minutes for 12 miles is a 2:50 per mile pace under a 35-pound ruck — faster than the world record for an unloaded 12 miles, so it is physically impossible. Action: return to the source record and correct it; if it cannot be corrected, exclude it and *document the exclusion*.

Record 19 (236 min) is unusual but legitimate. 236 minutes is 3 hours 56 minutes — slow, but entirely possible. Action: **keep it.** That is exactly the cadet the remedial-training decision is about; deleting it would bias the company's picture of its own risk.

Note the through-line to Question 9: 236 does not even clear the |z| > 2.5 screen. Statistics flag values for *review*; context decides.

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2 pts per record (classification and action). Full credit requires a contextual argument, not just "the number is big."

5. The modeling framework (4 pts)

- **Decision:** which cadets to place on the remedial ruck training plan before this term's march — a roster, not a number.
- **Objective:** maximize the number of cadets who earn a GO, subject to the limited remedial slots available.
- **Assumption (any one, with a reason):** last term's relationship between training volume and ruck time still holds; the recorded times are accurate and comparable (same route, load, weather); last term's cadets are representative of this term's.

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1 pt decision, 1 pt objective, 2 pts for an assumption that is genuinely required and explained.

Part II — Descriptive Modeling

56 POINTS

6. Mean and median (6 pts)

Sum = 1480. Mean = 1480 ÷ 8 = **185 minutes**

n = 8 is even, so the median averages the 4th and 5th ordered values: (175 + 180) ÷ 2 = **177.5 minutes**

The mean sits **7.5 minutes above** the median — the outlier at 236 doing work.

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3 pts mean, 3 pts median. Most common error: taking the 4th value (175) instead of averaging the middle two.

7. Sample standard deviation (8 pts)

x_i	168	172	175	175	180	184	190	236
$x_i - 185$	−17	−13	−10	−10	−5	−1	5	51
squared	289	169	100	100	25	1	25	2601

Σ = **3310** → s^2 = 3310 ÷ 7 = 472.86 → s = **21.75 minutes**

One observation contributes 2601 of 3310 — **79% of the total squared deviation from a single cadet**. That is why s is not resistant to outliers, and it motivates Question 8.

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2 pts squared deviations, 2 pts sum, 2 pts dividing by n − 1, 2 pts root and units. Dividing by 8 gives 20.34 — the single most common error.

8. Quartiles, IQR, and range (6 pts)

Lower half 168, 172, 175, 175 → Q_1 = (172 + 175) ÷ 2 = **173.5**

Upper half 180, 184, 190, 236 → Q_3 = (184 + 190) ÷ 2 = **187**

IQR = 187 − 173.5 = **13.5 minutes** Range = 236 − 168 = **68 minutes**

Interpretation: the range is five times the IQR. The middle half of the platoon is packed into a 13.5-minute band, so the 68-minute range is driven almost entirely by one slow cadet. Report the **IQR** as the spread of a typical cadet — it is resistant, and the range is not.

Grading note: Excel's `QUARTILE.INC` uses interpolation and returns **174.25** and **185.5** (IQR 11.25). Accept either set if the cadet names the method. The question specifies median-of-halves, so that is the primary answer.

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1 pt each for Q1, Q3, IQR, range; 2 pts for the interpretation, which must contrast resistance rather than just say "IQR is smaller."

9. z-scores and the outlier screen (8 pts)

(a) Using $\bar{x} = 185$ and $s = 21.75$ from Questions 6 and 7:
$z_8 = (236 - 185) \div 21.75 =$ **2.34** $\quad z_1 = (168 - 185) \div 21.75 =$ **−0.78**
Follow-on credit applies: a cadet who divided by n in Q7 gets s = 20.34 and $z_8 = 2.51$, which does clear the screen. Grade part (b) against their own s.

(b) **Neither observation is flagged.** The screen is $|z| > 2.5$; the slowest cadet reaches only 2.34, and the fastest is well inside at 0.78.

What the screen does and does not establish: it is a **rule of thumb for selecting values to review**, not a test of validity. Clearing it does not certify a value as correct, and exceeding it does not prove a value is wrong — Record 41 in Question 4 was an error that any screen would catch, while a plausible-looking mistyped 185 would pass every screen there is. Context, not the threshold, decides.

Direction also matters: a **positive** z on ruck time means **slower**, not better.

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4 pts (a), 2 per z-score with correct sign. 4 pts (b): 2 for correctly concluding neither is flagged, 2 for the explanation. A cadet who flags 236 has applied $|z| > 2$ instead of 2.5 — award the 2 explanation points if the reasoning is otherwise sound.

10. Choosing a measure of center (6 pts)

(a) **Median.** The distribution is clearly **right-skewed** — unimodal, with a long tail of slow times running out to 240 minutes and no matching tail on the fast side. The mean is pulled toward that tail, so it overstates what a typical cadet runs; the median is resistant to those extreme values.

(b) **Mean > median.** In a right-skewed distribution the long upper tail drags the mean above the median — consistent with 1st Platoon, where 185 > 177.5.

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4 pts (a): median selected AND justified with skew / outlier / resistance language. Median with a vague explanation earns 2. Mean selected earns 0 regardless of explanation. 2 pts (b).

11. Boxplot versus histogram (4 pts)

(a) The box edges are $\boldsymbol{Q_1 = 173.5}$ and $\boldsymbol{Q_3 = 187}$, and the line inside the box is the median, 177.5 — all three from Questions 6 and 8. They agree with the figure. The box width *is* the IQR, 13.5 minutes.

(b) The **boxplot** shows the five-number summary and applies the 1.5 × IQR fence, so it identifies the 236-minute cadet as a flagged point explicitly and makes the quartiles readable at a glance. The **histogram** shows the *shape* of the distribution — modality, where the bulk sits, how the tail decays — which the boxplot hides entirely. A boxplot cannot reveal a bimodal distribution; a histogram cannot give you the exact quartiles.

The right answer to “which is better” is that they answer different questions. For the commander's read-ahead, the histogram tells the story; for flagging individual cadets, the boxplot does.

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2 pts (a) — must name Q1 and Q3 specifically. 2 pts (b) — needs one genuine strength of each, not a restatement of what a boxplot is.

12. Reading a correlation table (8 pts)

(a) **Weekly training mileage**, $|r| = 0.92$. The **negative** sign means cadets who train more miles per week tend to post *lower* ruck times — they finish *faster*. A linear model on it is **reasonable** (0.92 is a strong linear association and a scatterplot would support a straight-line form) and **necessary** (the commander needs to predict a ruck time for cadets not yet measured, which a summary statistic cannot do).

(b) **No.** This is observational data; nobody assigned cadets to a training volume. A plausible third variable is **baseline fitness** — fitter cadets both choose to train more *and* ruck faster, which would produce this correlation with no causal effect of the training itself.

(c) **No.** Either reason earns full credit:

- **Restricted range.** Every cadet carried essentially the same 35 lb, so the variable barely varies and there is nothing for the statistic to detect. That does not make load irrelevant — it means this dataset cannot speak to it. Test it by varying the load.
- **Linearity.** r measures *linear* association only; a real but curved relationship can produce r near zero.

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3 pts (a) — 1 variable, 1 sign in context, 1 reasonable-and-necessary. 3 pts (b) — requires naming observational data or a confounder, not just reciting the slogan. 2 pts (c).

13. Write a caption (10 pts)

Model response. *Figure 1. Most of Company B rucks near 180 minutes, but a slow tail of roughly a dozen cadets runs past 200.* The histogram shows the distribution of 12-mile ruck times for all 64 cadets in Company B, binned in 10-minute intervals. The distribution is right-skewed: it peaks between 170 and 190 minutes, where more than half the company falls, then trails off through a long upper tail reaching 240 minutes. Because of that tail the company average (185 minutes) is slower than the typical cadet's time (177.5 minutes), so quoting the average alone understates how most of the company performs and hides the group actually at risk. The decision is not about the company average — it is about the dozen or so cadets past 200 minutes, and a remedial program should be sized to them.

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10 — Dominates. *Informative title; fully describes the figure (variable, units, n, shape); gives a key takeaway and its significance for this audience, using evidence from the figure.*
8 — Proficient. *Caption elements incomplete or slightly inaccurate; takeaway imprecise or not fully supported.*
6 — Marginal. *Significant omissions or mathematical inaccuracies; takeaway does not answer the commander's question.*
4 — Marginal. *Clear misunderstanding of the figure's elements.* ***0.*** *Blank. (−1 minor error.)*

Where cadets lose points: no title; describing the axes but never stating a takeaway; writing for a statistician instead of for the commander who has to decide something.

Part III — Linear Regression and Residuals

50 POINTS

14. Read the line of best fit off the graph (6 pts)

(a) At $x = 10$ the line reads **198**; at $x = 30$ it reads **174**. Accept 197–199 and 173–175.

(b) $b_1 = (174 - 198) \div (30 - 10) = -24 \div 20 =$ **-1.2**; $b_0 = 198 + 1.2(10) =$ **210**; **$\hat{y} = 210 - 1.2x$**

Two habits: read the **line**, never a nearby data point; and put rise over run in that order — inverting gives −0.83, the usual slip.

2 pts readings, 2 pts slope (accept −1.1 to −1.3), 2 pts equation with a correctly back-solved intercept.

15. Interpretation (8 pts)

(a) Each **additional mile per week** of training is associated with a **1.2-minute decrease** in predicted 12-mile ruck time. Say *associated with*, not *causes* — observational data.

(b) The intercept says a cadet who trains **0 miles per week** is predicted to ruck 12 miles in **210 minutes**. **Not meaningful here**: the training data covers only 5 to 30 mi/wk, so $x = 0$ is **extrapolation outside the range of the data**.

4 pts (a): number, both units, direction, association language. "The slope is −1.2" alone earns 1. 4 pts (b): 2 literal reading, 2 extrapolation.

16. R-squared (4 pts)

Approximately **85.4%** of the variation in **12-mile ruck time** is explained by its linear relationship with **weekly training mileage**. Describe it as **strong**.

What it does not tell you: that the linear model is the *right shape* (Q18 shows it is not), that the model predicts well **out of sample** (Q20 shows it degrades), or that mileage causes the difference. R^2 is computed on the **training data only**.

2 pts for the completed sentence — "the model is 85% accurate" is wrong and earns 0. 1 pt strong. 1 pt for a correct limitation.

17. Residuals and training error (8 pts)

(a) $x = 5$: $\hat{y} =$ **204**, $e =$ **+6** $x = 15$: $\hat{y} =$ **192**, $e =$ **−3** $x = 30$: $\hat{y} =$ **174**, $e =$ **+6**

(b) Residuals +6, −3, −3, −3, −3, +6 → MAE = 24 ÷ 6 = **4.0 minutes**. On the data it was fit to, the model misses by about 4 minutes on average.

(c) The signed residuals sum to **zero** — a property of the least-squares line, not a coincidence — so their average is 0 and would report a perfect model no matter how badly it fit. Absolute values stop positive and negative misses from cancelling.

3 pts (a), 3 pts (b), 2 pts (c). Sign errors on e cost 1 pt once, not per row.

18. Residual plot and the regression assumptions (10 pts)

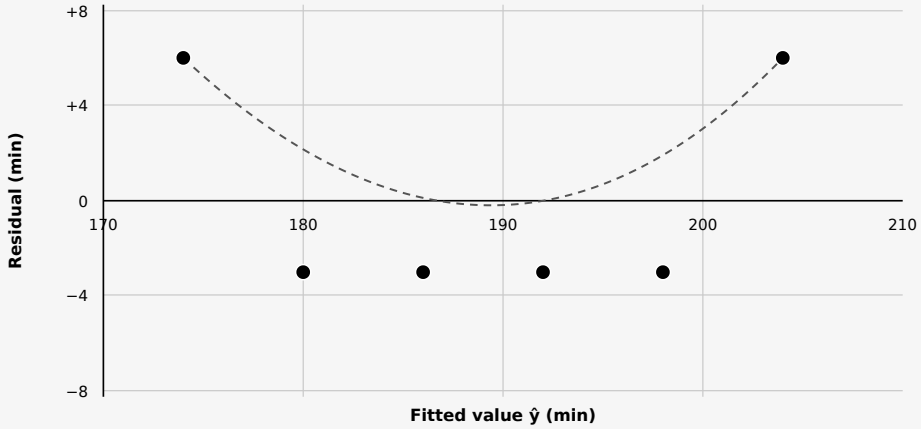


Figure 4 (key) — points at ŷ = 174, 180, 186, 192, 198, 204 with residuals +6, −3, −3, −3, −3, +6.

Plotting note: because the slope is negative, the largest fitted value (204) belongs to the *smallest* x (5 mi/wk). Cadets who carry over their Q17 table in x-order will place the points left-to-right backwards. The U-shape survives either way, but the axis is ŷ.

Linearity — violated. The residuals form a clear **U-shape**: positive at both ends, negative through the middle. They are not randomly scattered about $e = 0$. The line systematically over-predicts in the middle of the range and under-predicts at both ends; a predictable error is a modeling failure, not noise. **Next step:** fit a nonlinear trendline — a quadratic for a single bend — and compare on the **test set**, not on R^2.

Constant variance — no evidence of a violation. The residuals sit at magnitudes 6, 3, 3, 3, 3, 6; the vertical spread neither fans out nor compresses systematically as ŷ increases. With only six points this is a weak read, and cadets should say so.

Outliers — none. No residual sits far outside the range of the others; the two +6 values are the extremes but are the same size as each other and are part of the pattern, not isolated from it.

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4 pts plot (all six points placed against ŷ), 2 pts linearity, 2 pts constant variance, 2 pts outliers. Full credit requires citing evidence from the plot, not reciting definitions. "The points are close to zero so the model is good" earns 0 on linearity — the pattern, not the size, is the finding. Award constant-variance credit to any cadet who notes the sample is too small to judge confidently.

19. Distribution of residuals (4 pts)

Mean = **0.00 min** Standard deviation = **4.65 min** Min = **−3 min** Max = **+6 min**

Work for the standard deviation: the residual mean is 0, so $\Sigma e^2 = 36 + 9 + 9 + 9 + 9 + 36 = 108$; $s = \sqrt{(108 \div 5)} = \sqrt{21.6} = 4.6476$.

The mean is **exactly** zero, as it must be for a least-squares fit — that is a property of the method, so it confirms the arithmetic but says nothing about whether the model is any good. A mean substantially different from zero would mean the line was **not** the least-squares line: an arithmetic error, a model fit on different data than it is being scored on, or a trendline forced through the origin.

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1 pt per statistic with units (min and max may be given unsigned only if direction is stated). 1 pt bonus folded into the explanation: full credit requires recognizing that mean = 0 is guaranteed, not evidence of quality.

20. Test error versus training error (10 pts)

x	y	$\hat{y}$	$\lvert y - \hat{y} \rvert$
8	208	200.4	7.6
18	183	188.4	5.4
28	185	176.4	8.6
Test MAE = 21.6 ÷ 3			**7.20**

Test MAE = 7.20 minutes against a training MAE of 4.00 — the model is **80% worse on cadets it has never seen**.

Model answer: “On the cadets used to build it, the model predicts ruck time to within about 4 minutes. On cadets held out of the fit it is off by about 7 minutes, so 7 minutes — not 4 — is the honest accuracy to plan around. Some gap is normal because the fit is tuned to the training cadets; a gap this large also matches the curved residual pattern, which says the straight line is the wrong shape. Usable for coarse triage, but I would not use it to separate a cadet predicted at 178 minutes from one predicted at 183.”

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6 pts table (2 per row), 1 pt MAE. 3 pts interpretation: 1 for test error being the higher and honest number, 1 for noting some degradation is expected, 1 for a consequence in the commander's terms.

> **Watch for:** cadets who average the *signed* errors (7.6 − 5.4 + 8.6 = 10.8, ÷3 = 3.6) and conclude test error is lower than training error.

Part IV — K-Nearest Neighbors

32 POINTS

21. Set up the model (6 pts)

(a) The threshold is **180 minutes**, the Army standard for the 12-mile foot march: ruck time ≤ 180 is a GO. It is defensible because it is the **decision boundary that already exists in the real problem** — the commander's question is literally “who will pass,” so the class label matches the decision. Splitting at the median instead would force exactly half the company into each class by construction, producing a label that is an artifact of this sample rather than a fact about the standard, and the resulting model would answer a question nobody asked.

(b) The **training** mean and standard deviation define the scale the model was built on, so the test records must be mapped onto that same scale to be comparable; re-standardizing with test statistics moves the yardstick between fitting and scoring, and the distances would no longer mean the same thing. It also **leaks information** from the test set into the modeling process, which destroys the test set's one job: being an honest stand-in for data the model has never seen. In deployment you would have no test set to compute statistics from anyway — a single new cadet has no mean.

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3 pts (a) — 1 for the threshold, 2 for tying it to the decision rather than to the data. 3 pts (b) — 2 for the same-scale / comparability argument, 1 for leakage or the deployment argument.

22. Standardize (8 pts)

(a) Sum = 8 + 16 + 20 + 20 + 24 + 32 = 120 → $\bar{x}$ = **20 mi/wk**

Deviations −12, −4, 0, 0, +4, +12; squares 144, 16, 0, 0, 16, 144; sum = 320 → s = √(320 ÷ 5) = √64 = **8 mi/wk**

(b) z_1^* = (30 − 20) ÷ 8 = **1.25** z_2^* = (12 − 14) ÷ 2 = **−1.00**

Read Q off those numbers: trains 1.25 SD more than typical, runs 1.00 SD faster than typical.

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2 pts mean, 3 pts s (1 of 3 lost for dividing by 6), 3 pts both z-scores with correct signs.

23. Standardized distances (8 pts)

Cadet	z_1	z_2	Δz_1	Δz_2	d	Rank	Class
A	−1.50	0.50	−2.75	1.50	3.13	6	NO-GO
B	−0.50	0.00	−1.75	1.00	2.02	4	NO-GO
C	0.00	1.50	−1.25	2.50	2.80	5	NO-GO
D	0.00	−0.50	−1.25	0.50	1.35	3	NO-GO
E	0.50	−1.50	−0.75	−0.50	0.90	1	GO
F	1.50	0.00	0.25	1.00	1.03	2	GO

Sample calculation, nearest cadet E:
$d = \sqrt{[(0.50 - 1.25)^2 + (-1.50 - (-1.00))^2]} = \sqrt{[(-0.75)^2 + (-0.50)^2]} = \sqrt{0.8125} =$ **0.90**

Neighbor order: E, F, D, B, C, A.

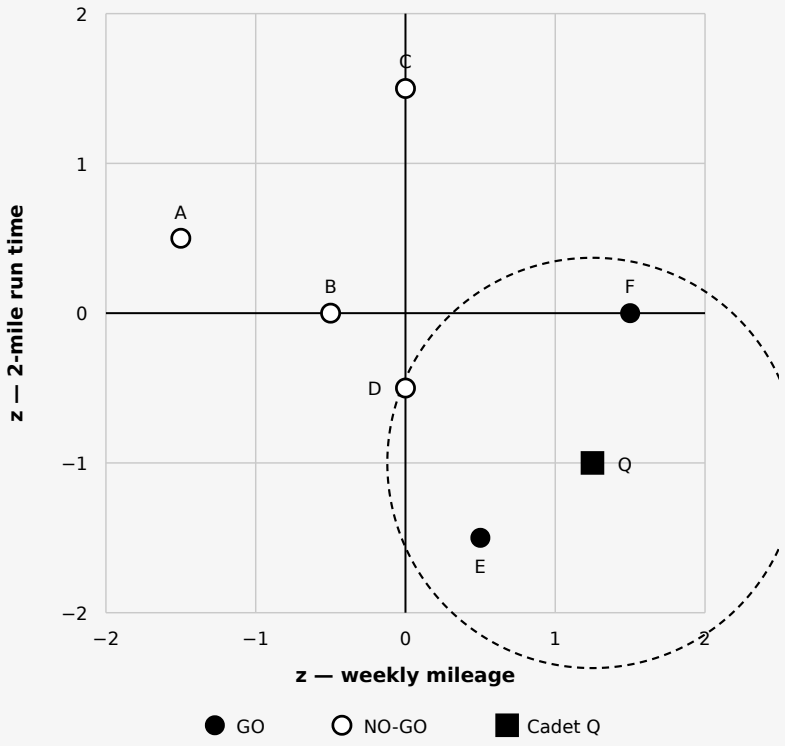


Figure 5 (key) — standardized space. The dashed ring is the k = 3 neighborhood of Q: E, F, D.

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3 pts standardized columns, 4 pts distances, 1 pt ranking. Follow-on credit: a wrong s from Q22 carried correctly through here still earns the distance points.

> **Watch for:** forgetting to square the difference, or subtracting distances instead of coordinates. Also −1.50 − (−1.00) = −0.50, not −2.50.

24. Choose k and classify (6 pts)

(a) n_{train} = **6**; √6 ≈ **2.45**; nearest odd integer = **k = 3**

Why odd: with two classes, an even k can split the vote evenly (2–2, 3–3) and leave the classifier with no majority and no principled way to break the tie. An odd k guarantees a strict majority.

(b)

k	Neighbors	Vote	Predicted class
1	E	1 GO – 0 NO-GO	**GO**
3	E, F, D	2 GO – 1 NO-GO	**GO**
5	E, F, D, B, C	2 GO – 3 NO-GO	**NO-GO**

The prediction **flips** between k = 3 and k = 5. That instability is itself a finding and sets up Question 25.

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3 pts (a) — 1 for k = 3, 2 for the tie-breaking explanation. 3 pts (b), 1 per row; the vote count must be shown, not just the class.

25. Sensitivity to k (4 pts)

Accuracy: **k = 1 → 3/4 = 75%** **k = 3 → 4/4 = 100%** **k = 5 → 2/4 = 50%**

Use k = 3. It has the highest test accuracy, and it is also what the $\sqrt{n_{\text{train}}}$ rule of thumb produced independently — the rule and the evidence agree.

What goes wrong at k = 1: the prediction rests entirely on one neighbor, so a single atypical training record decides the answer. Cadet T4 is the demonstration — its nearest neighbor is F, a GO, so k = 1 calls it GO, but T4 is really a NO-GO. This is **overfitting**: k = 1 reproduces the training data perfectly (each point is its own nearest neighbor, so training error is always 0) while generalizing worse.

What goes wrong at k = 5: the neighborhood grows so large it stops being local — it reaches across the whole feature space and washes out the structure near the target. Worse, the training set holds only **2 GO cadets out of 6**, so at k = 5 a GO majority is **arithmetically impossible**: at most 2 of any 5 neighbors can be GO. The model can no longer predict the minority class at all, which is why it misses both true GO cadets.

Extending that: at **k = 6 = n**, every cadet gets the same answer — NO-GO, the majority class — regardless of their features. That is the classification analogue of the mean-baseline model in Question 26.

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2 pts accuracies (all three). 2 pts explanation, 1 for the k = 1 overfitting argument and 1 for the k = 5 over-smoothing argument. A cadet who notes the minority-class impossibility at k = 5 has earned the point outright.

Part V — Synthesis

8 POINTS

26. Compare all models (5 pts)

Model	R^2	Train MAE	Test MAE	Test accuracy
Baseline (mean, 189 min)	N/A	9.00	9.67	N/A
Linear regression	0.854	4.00	7.20	N/A
Degree-5 polynomial	1.000	0.00	14.50	N/A
KNN classification, k = 3	N/A	N/A	N/A	100%

Why the N/A cells: R^2 and MAE measure error in the *units of a continuous response*; KNN here predicts a category, and there is no meaningful distance between GO and NO-GO to average. Conversely the three regression models produce a number of minutes, not a class, so accuracy does not apply unless you first threshold their predictions at 180.

Recommendation. Among the models predicting ruck time, **linear regression**: selection is made on **test** MAE, the only column estimating performance on unseen cadets, and 7.20 beats the do-nothing baseline of 9.67. The **degree-5 polynomial is the trap** — R^2 = 1.000 and training MAE = 0.00 look perfect because a degree-5 curve has exactly enough flexibility to pass through six points, fitting the noise along with the signal. Its test MAE of 14.50 is the worst of the three, worse than predicting the mean. That is **overfitting**, and it is the direct answer to HW2's question about whether a model can have a higher R^2 and a worse test error: yes, and this is how.

A cadet who argues for **KNN** instead is also correct if they note it answers the commander's actual question — who passes — rather than predicting a time and thresholding it afterward. Award full credit for that reasoning.

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2 pts filled table including correct N/A placement, 1 pt selecting on test MAE, 2 pts diagnosing the polynomial as overfitting. Selecting the polynomial earns 0. Selecting linear regression "because R^2 is high" earns 1 — right answer, wrong criterion.

27. Contrast the two model families (3 pts)

Linear regression is a global, parametric model. It commits in advance to a shape — a straight line — and compresses the entire training set into two numbers, the slope and the intercept. Once fitted, the training data is discarded; a prediction is an arithmetic evaluation of the equation. It assumes the relationship really is linear everywhere in the range, and when that assumption fails you see it in the residual plot.

KNN is a local, non-parametric model. It assumes no functional form at all. It keeps every training record and, at prediction time, finds the records nearest the target and lets them vote. Its only assumption is that **nearby points behave similarly** — which is why the distance metric, and therefore standardization, is doing all the work.

The trade: regression extrapolates (with the risks of Q15b) and gives an interpretable slope; KNN adapts to any shape but cannot extrapolate beyond its neighbors, has no coefficient to interpret, and degrades when features are on mismatched scales or when the training set is small.

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3 pts: 1 for the global-equation versus local-neighbors mechanism, 1 for regression assuming a functional form, 1 for KNN assuming only that near points are similar. A cadet who only says "one uses a line and one uses neighbors" earns 1.

Running the Session

SUGGESTED THREE-AND-A-HALF-HOUR BLOCK

The point of the timed run is not the score. It is to find out which of the seven failure modes on page 1 each cadet owns, while there is still a day to fix it.

Time	Activity
0:00 – 0:10	**Cold-start drill.** Five numbers on the whiteboard. Everyone computes mean, median, and s by hand. Anyone who divides by n instead of $n - 1$ gets caught here rather than on the WPR.
0:10 – 1:55	**Timed run.** 105 minutes, calculators only, phones down. Call the part transitions out loud at 14, 46, 78, and 100 minutes so cadets learn the pacing.
1:55 – 2:15	**Self-scoring.** Hand out this key and have cadets score their own paper against the rubrics. Self-scoring beats hearing you read answers — they have to argue with the rubric.
2:15 – 3:10	**Board the four that break people:** Q7 ($n - 1$), Q18 (residual pattern against *fitted values*), Q20 (train versus test MAE), Q22–25 (standardize, distance, majority vote, sensitivity to k). Work each at the board with a cadet driving.
3:10 – 3:30	**Round the room.** Every cadet names the one thing they will review tonight. Write the list down — it tells you what to push to the whole company.

If you are short on time

Cut to a **50-minute core**: Questions 6, 7, 8, 9, 14, 17, 20, 22, 23, 24. That is 74 points and covers every computation the class has flagged as likely — mean, standard deviation, quartiles, z-scores, line of best fit, residuals, training versus test MAE, and the full KNN chain through the majority vote. Assign the interpretation items as overnight work; those are the points cadets most often leave on the table and they can be reviewed without a calculator.

Overnight reps — drawn from HW1 and HW2, not on this practice test

Cadets who want more repetitions should work these directly out of the homeworks, using Penguins, Ames Housing, or the Happiness Report as the dataset:

- **Composite z-scores** (HW1 Q18). Standardize an observation on two variables and sum them, negating the z-score of any variable where *lower is better*. Ruck time versus weekly mileage is the natural pair here.
- **Correlation assumptions** (HW1 Q22): both variables numerical, scatterplot supports a linear form, no outliers distorting r.
- **Nonlinear trendlines** (HW2 §4). The U-shaped residuals in Q18 say a quadratic should be fitted and compared on test MAE. Working this end to end is the single best use of an evening.
- **The 80/20 partition and the frozen random column** (HW2 Q3) — conceptually, why the random numbers must be frozen before the split.
- **Out-of-sample prediction at a chosen x** (HW2 Q10c), and whether that prediction is practically reasonable.

Scoring bands (out of 170)

Score	Read
153–170	Ready. Have them help teach the review block.
130–152	Solid on computation; losing interpretation points. Drill the “in context, with units” habit.
102–129	One or two systematic errors, usually $n - 1$ or train-versus-test. Find the specific one and fix it.
Below 102	Needs a full walk-through of Parts III and IV before Monday. Pair with a cadet in the top band.

MA103 Data-Driven Modeling · AY27-1 · Unofficial practice material prepared for the WPR 1 study session.
Coverage drawn from the published lesson objectives for Lessons 1–21 and from Homeworks 1 and 2. Not the examination.